

System Capabilities

Live, simulated, and deliberately not built — drawn from the system's own truth table, not from a deck.



LIVE

Pre-submission claim integrity for NPHIES

Ambient documentation, a clinician-reviewed SOAP note, coded orders and claim readiness in one loop — with every claim traceable back to the record it was built from.

- Deterministic facts: the model formats text, it never decides a fact, a code or a dose
- The clinician signs every clinical action — about seven explicit approvals on the headline path
- Evidence links must entail: a citation supports its claim or it does not ship

When a fact or a control is missing the system refuses or defers and says so, rather than guessing: the pilot establishes the local rates, not a borrowed benchmark.



LIVE

Read from the running system

0 SFDA-coded medications in the seed
0 contraindication edges loaded
1 renal-dose rule loaded

So the necessity and interaction checks defer instead of turning green. Unknown is not green.



LIVE

Ambient Documentation

- Live dictation, on-prem
- SOAP drafted; the model only formats
- Checklist quotes are verified
- Draft save and sign, auditable
- Demo playback is scripted



LIVE

Grounded Q&A

- Classify, then answer or refuse
- Retrieves the PostgreSQL index
- Blocklist gate fails closed
- A refusal still offers the facts
- Citations carry the row id



LIVE

Evidence Chain

- Reasons over the patient graph
- Shows the query it really ran
- Interaction and renal modules run
- Necessity uses pre-auth rules



LIVE

Claims & Coding

- Deterministic readiness checks
- ICD/SBS from the governed map
- Coder queue is restart-safe
- Green only when the payer is done
- Payer and HIS replies are stubs



LIVE

Governance & Audit

- SHA-256 audit chain with a verify endpoint
- Audit writes land in a durable outbox
- The WORM export reads its object back
- Reference data is checksummed before use
- Erasure is a workflow that records proof



LIVE

Access & Scope

- Keycloak OIDC login on the dev realm
- RBAC guard on every guarded route
- Patient scope: out of scope is 403
- Revocation checked on every request
- Sessions, OTP limits and OIDC state shared



LIVE

Platform

- Core :4000, web :3000, six AI services :5001-5006
- Postgres, Neo4j, Keycloak, MinIO, Redis in Docker
- /health, /health/ready and /health/preflight
- The ingestion sweep holds an advisory lock
- Containers for core and web are the L01 item



SIMULATED

Connector Modes — What Is Real

- Model provider: deepseek — a real cloud API
- Transcription: local faster-whisper large-v3
- NPHIES: a stub. There is no real payer behind it
- HIS: a stub. There is no hospital system behind it
- SMS/OTP: a stub. The code is logged, never sent

Every mode is printed on /health/ready, so no one has to infer which parts of this are real.



NOT BUILT

Not Built, Not Claimed

- Real patient data — every patient is synthetic
- Live NPHIES submission, eligibility or pre-auth
- A hospital FHIR feed
- Arabic ASR validation, and Arabic dictation
- An independent evaluation set with adjudication

No borrowed benchmark and no ROI figure appears in any claim this system makes.

THE FIVE NON-NEGOTIABLES

- 01 Claims match code
- 02 Facts are deterministic
- 03 Unknown is not green
- 04 Measured locally or silent
- 05 The clinician signs

SOURCES THIS MAP WAS DRAWN FROM

docs/DEMO_TRUTH_TABLE.md

docs/assessment/PRE_DEMO_READINESS_ASSESSMENT.md

live Postgres + Neo4j checks, 18 Sep 2026

Synthetic data only · not for clinical use · every status is traceable to a file, a query or an endpoint